The solow model

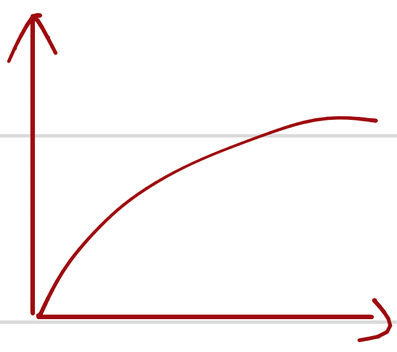
1. Assumption

(1) Assumption on Production.

① $Y = F(K, AL)$.

(a) $F(\lambda K, \lambda AL) = \lambda F[K, AL]$. $\forall \lambda > 0$

(b) $\frac{\partial F}{\partial K} > 0$ $\frac{\partial^2 F}{\partial K^2} < 0$; $\frac{\partial F}{\partial L} > 0$ $\frac{\partial^2 F}{\partial L^2} < 0$.



(c) $\lim_{K \to 0} \frac{\partial F}{\partial K} = \infty$ $\lim_{K \to \infty} \frac{\partial F}{\partial K} = 0$

$\lim_{L \to 0} \frac{\partial F}{\partial L} = \infty$ $\lim_{L \to \infty} \frac{\partial F}{\partial L} = 0$

> Inada condition.

(d) $\lim_{L \to \infty} F(K, AL) = \infty$ $\lim_{K \to \infty} F(K, AL) = \infty$

② $F(\frac{K}{AL}, \frac{AL}{AL}) = F(\frac{K}{AL}, 1) = \frac{1}{AL} F(K, AL)$

设 $k = \frac{K}{AL}$ $L = \frac{L}{AL}$ $f(k) = F(k, 1) = y$.

把 F 写成 $AL \cdot f$ ：把 k 写成 $\frac{K}{AL}$

so MPK $= \frac{\partial F(K, AL)}{\partial K} = \frac{\partial [AL f(\frac{K}{AL})]}{\partial K} = AL \cdot \frac{1}{AL} f'(\frac{K}{AL}) = f'(k) > 0$.

MPL $= \frac{\partial F(K, AL)}{\partial L} = \frac{\partial [AL f(\frac{K}{AL})]}{\partial L} = A f(\frac{K}{AL}) + AL f'(\frac{K}{AL}) \cdot \frac{K}{A} \cdot \frac{1}{-L^2}$

$= A f(k) - A k f'(k)$

$= A[f(k) - f'(k) \cdot k] > 0$ ：

Cobb-Douglas function

$F(K, AL) = K^a (AL)^{1-a}$ $0 < a < 1$

$MPK = \frac{\partial F}{\partial K} = a K^{a-1} (AL)^{1-a}$. $MPL = \frac{\partial F}{\partial L} = (1-a) K^a A^{1-a} L^{-a}$

So $MPK \cdot K + MPL \cdot L = Y$.

$\frac{MPK \cdot K}{Y} = \alpha$ $\frac{MPL \cdot L}{Y} = 1-\alpha$

② $\dot{A}(t) = g A(t)$ $\dot{L}(t) = nL(t)$. , $A(0)$, $L(0)$

A的技术变化率是 g (常数)

δ. Is given.

Can be written $\frac{\dot{A}}{A} : g$ $\frac{\dot{L}}{L} = n$.

Also we know $\dot{x}(t) = \frac{dx(t)}{dt}$.

So

(2) Assumption on household.

s as exogenous 给定的.

2. Behaviour.

(1) Behaviour of the household.

① Saving rate = s.

$I = sY$ $C = (1-s)Y$.

(2) Behavior of the Firms.

① $\max F(K, AL) - RK - wL$.

rent price

FOCs: $F_K(K, AL) = R$

$F_{AL}(K, AL) = w$

" $AF_2(K, AL)$

K的动态过程 (2 ways)

② $\dot{k} = I - \delta k = sY - \delta k = sF(K, AL) - \delta k$.

投资 折旧

方法1.

$\Rightarrow \frac{\dot{k}}{k} = \frac{sF(K, AL)}{k} - \delta$

$\because K = AL \cdot k$ So $\frac{\dot{K}}{K} = \frac{\dot{A}}{A} + \frac{\dot{L}}{L} + \frac{\dot{k}}{k}$ → 乘法求导法则 ④.

$= g + n + \frac{\dot{k}}{k}$.

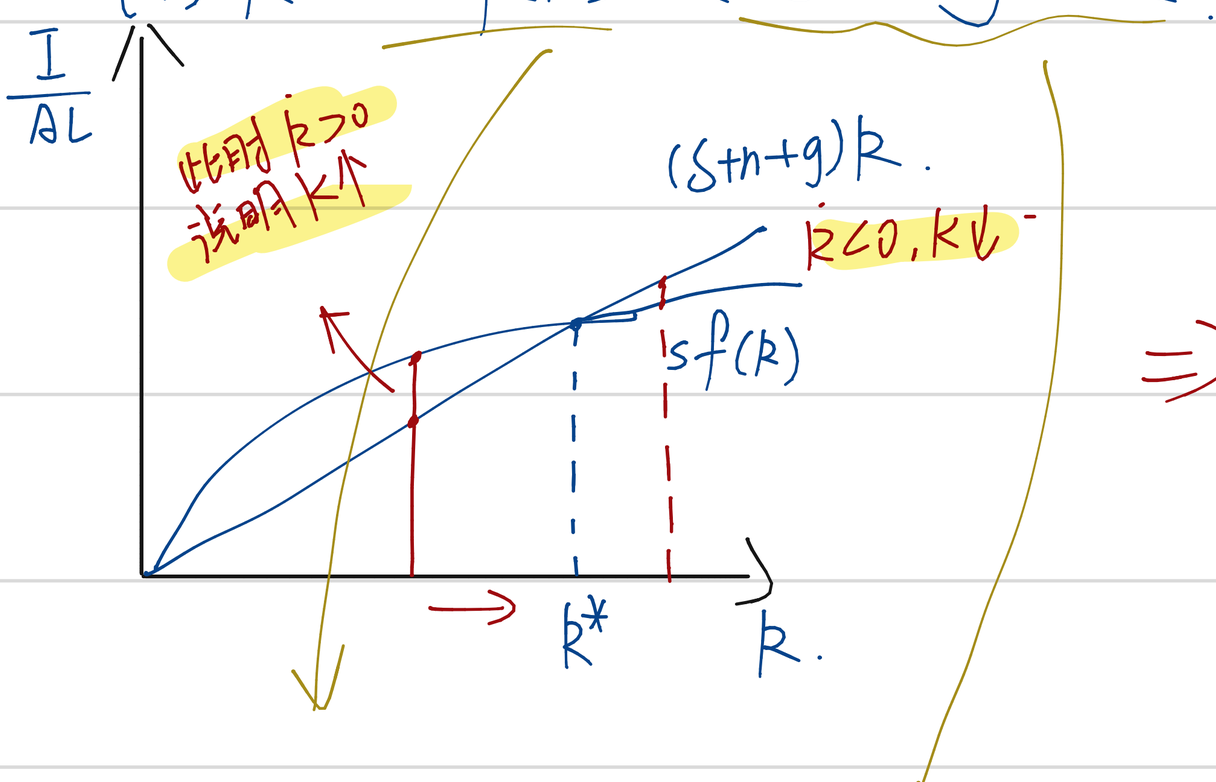
So $n+g+\frac{\dot{k}}{k} = \frac{sAL \cdot f(k)}{K} - \delta$

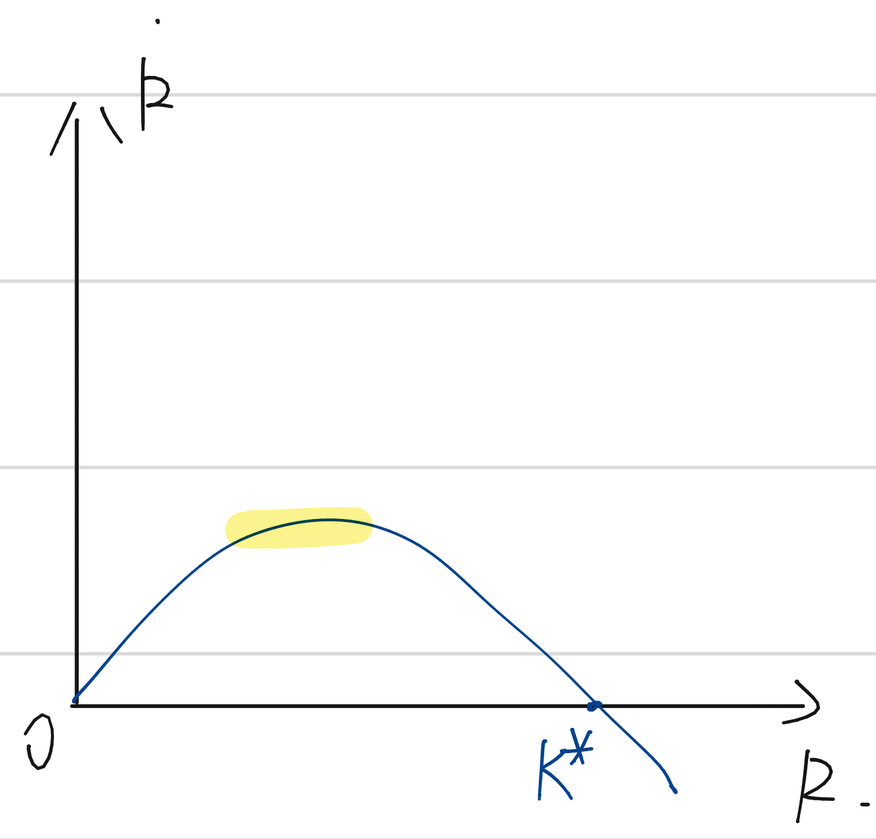
$\frac{\dot{k}}{k} = \frac{sf(k)}{k} - (\delta+n+g)$.

or $\dot{k} = sf(k) - (\delta+n+g)k$

3. Equilibrium.

(1) $\dot{k} = sf(k) - (\delta+n+g)k$.





经济含义：人均资本增加量 — 为了使人均资本不减少 所必须增加的量.

数理 (法2)

$\dot{k}=0 \Rightarrow sf(k^*) - (\delta+n+g)k^* = 0$ (*)

$\therefore sf(k^*) = (\delta+n+g)k^*$

$\frac{f(k^*)}{k^*} = \delta+n+g$

$\left[\frac{f(k)}{k}\right]' = \frac{k \cdot f'(k) - f(k)}{k^2} = -\frac{f(k) - kf'(k)}{k^2} < 0$

上导下不导减去上不导下导，除以下的平方. ☆

洛必达：上下各自求导，再求极限.

$\lim \frac{f(x)}{g(x)} = \lim \frac{f'(x)}{g'(x)}$

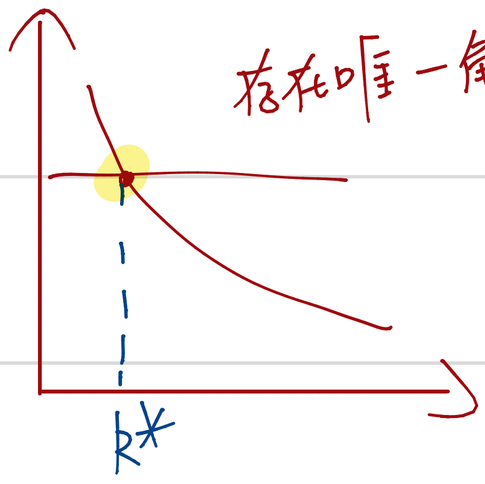
and $\lim_{k\to 0} \frac{f(k)}{k} = \frac{f'(k)}{1} = \infty$

Inada condition

$\lim_{k\to\infty} \frac{f(k)}{k} = \lim_{k\to\infty} \frac{f'(k)}{1} = 0$

Using the intermediate value theorem

there must exist $k=k^*$. let $\frac{f(k^*)}{k^*} = \frac{\delta+n+g}{s}$.



and because $\left[\frac{f(k)}{k}\right]' < 0$, so $\frac{f(k)}{k}$ is monotone decreasing

so k^* is unique.

BGP = Balanced Growth Path.

growth rate of output per worker determined by the rate of technological progress.

(2) Growth rate at the steady state

$$\frac{\dot{K}}{K} = \frac{\dot{k}}{k} + \frac{\dot{A}}{A} + \frac{\dot{L}}{L} = 0 + n + g = n + g.$$

$$\frac{\dot{Y}}{Y} = \frac{\dot{y}}{y} + \frac{\dot{A}}{A} + \frac{\dot{L}}{L} = \frac{\dot{k}}{k} + \frac{\dot{A}}{A} + \frac{\dot{L}}{L} = n + g.$$

↘ ∵ y 是 k 的函数. 见 Assumption ②.

$$\frac{\dot{C}}{C} \quad C = (1-s)Y \quad \frac{\dot{Y}}{Y} = n + g$$

$(K/L)/(K/L) = d[\ln(K/C)]/dt = g = (\dot{Y}/L)/(Y/L)$

一个经济体人均资本增速 = 人均产出增速 g.

人均GDP增速 − 人口增速 = 国家技术进步的增速.

k 和 y

(3) The speed of convergence 收敛速度.

$$\dot{k} = \dot{k}\big|_{k=k^*} + \frac{d\dot{k}}{dk}\Big|_{k=k^*}(k-k^*)$$

$\dot{k}|_{k=k^*} = 0$

在…处取值

$$= \frac{d\dot{k}(k)}{dk}\Big|_{k=k^*} \quad (k-k^*).$$

$$\lambda = -\frac{d\dot{k}(k)}{dk}\Big|_{k=k^*} \qquad \dot{k} = sf(k) - (\delta+n+g)k.$$

$$= -[sf'(k) - (\delta+n+g)]\Big|_{k=k^*}$$

$$= -[sf'(k^*) - (\delta+n+g)]$$

$$= -\left[\frac{(\delta+n+g)k^*}{f(k^*)} f'(k^*) - (\delta+n+g)\right]$$

$s = \frac{(\delta+n+g)k^*}{f(k^*)}$.

MPk.

$$= (\delta+n+g)\left[1 - \frac{k^* f'(k^*)}{f(k^*)}\right]$$

$\because k \cdot MPk = \partial k$

$$= (\delta+n+g)[1 - \partial k(k^*)] \text{ 常数} \cdot (\lambda).$$

$\because k^*$ 是常数 $\therefore \frac{d(k-k^*)}{dt} = \dot{k}$.

And $\dot{k} = -\lambda(k-k^*) \Rightarrow \frac{d(k-k^*)}{dt} = -\lambda(k-k^*)$

令 $x = k - k^*$

$\therefore \frac{dx}{dt} = -\lambda x$

两边积分 $\frac{dx}{x} = -\lambda dt$.

$\ln x = -\lambda t + C$.

两边取指数

$$k - k^* \simeq e^{-\lambda t}[k(0) - k^*].$$

$$x = e^{-\lambda t + c}$$

$$= e^{c} \cdot e^{-\lambda t}$$

令 $A = e^{c}$

$\therefore x(t) = A \cdot e^{-\lambda t}$

当 $t=0$ 时, $x(0) = A$

$\therefore A = x(0) = k(0) - k^*$

代回去.

$$k(t) - k^* = e^{-\lambda t}\left[k(0) - k^*\right]$$

So for $y = f(k)$ → 泰勒展开.

$$f(x) \approx f(a) + f'(a)(x-a) + f''(a)/2!\cdot(x-a)^2 + \cdots$$

$$y \simeq f(k^*) + f'(k)(k-k^*) \text{ 一阶展开.}$$

$$y - y^* \simeq f'(k^*)(k-k^*)$$

$$\ln(y-y^*) \simeq \ln f'(k^*) + \ln(k-k^*)$$

constant

$$\Rightarrow \frac{d[\ln(y-y^*)]}{dt} \approx \frac{d[\ln(k-k^*)]}{dt}.$$

$$\Rightarrow \frac{(\dot{y}-\dot{y}^*)}{y-y^*} \approx \frac{(\dot{k}-\dot{k}^*)}{k-k^*}.$$ same with k.

So $y - y^* \simeq e^{-\lambda t}\left[y(0) - y^*\right]$.

$$\frac{k(t)-k^*}{k(0)-k^*}$$

指数缩短 cy 类先验.

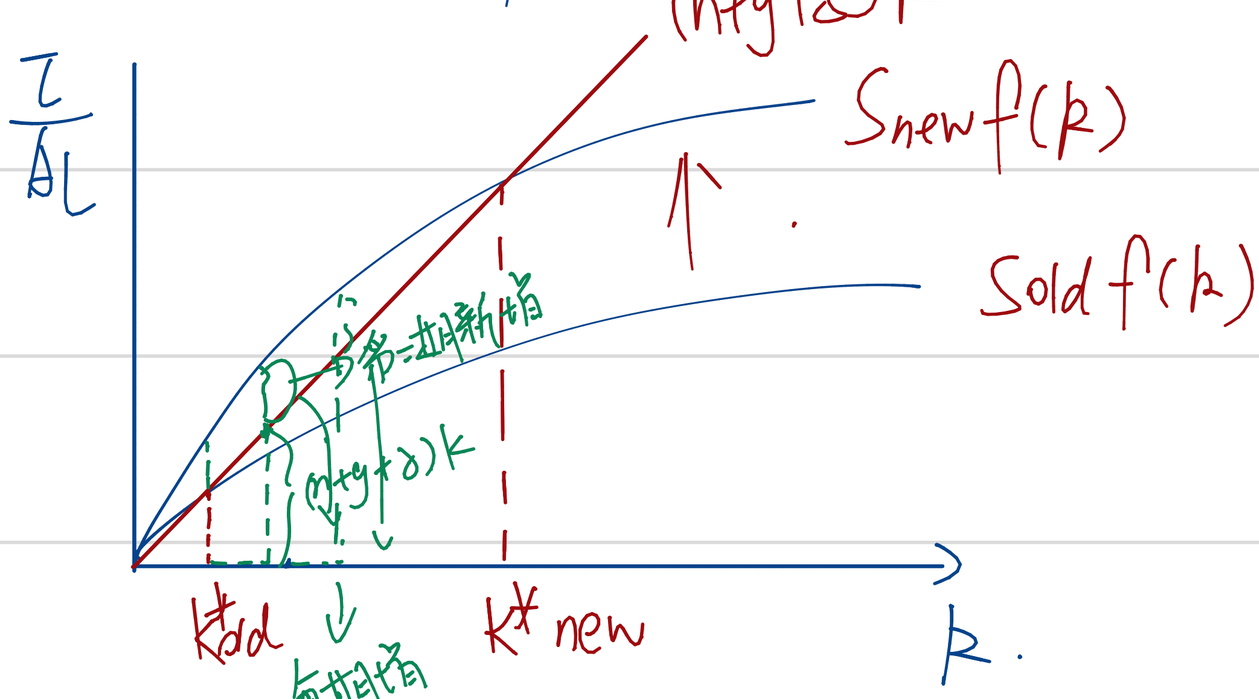
4. Shock.

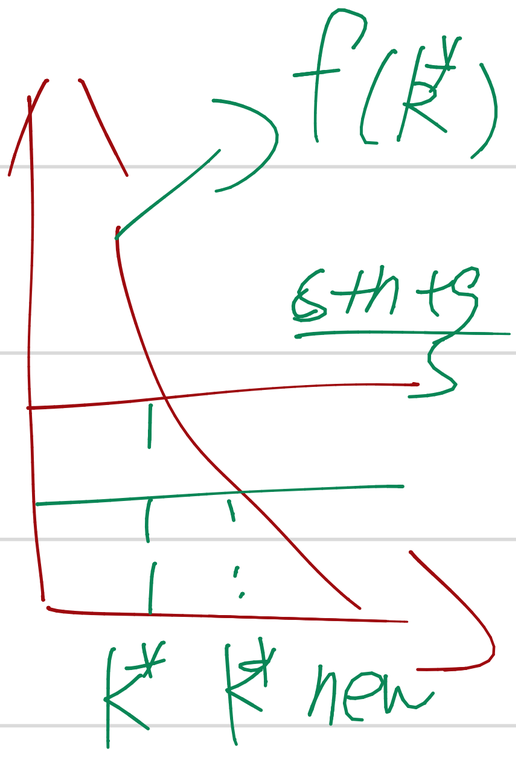
(1) A change in the saving rate

① s↑. permanent.

(n+g+δ)k.

s↑ → k*↑.





Because $\dfrac{f(k^*)}{k^*} = \dfrac{\delta+n+g}{s}$.

右移. k*↑.

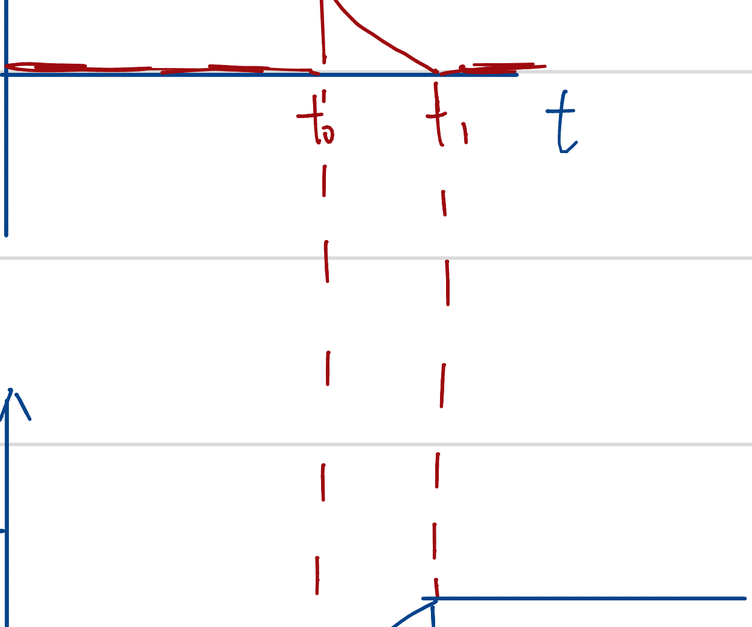
and $g(k) = \frac{f(k)}{k}$ ↓ is monotone decreasing.

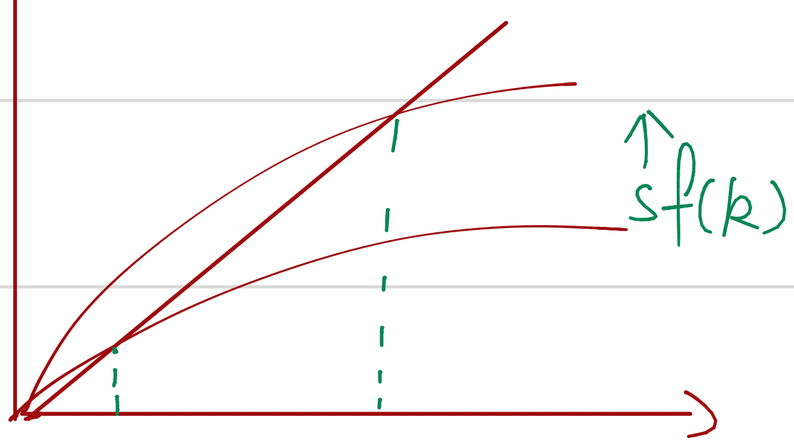
$s\uparrow$: $s_{old} < s_{new} \Rightarrow \frac{f(k^*)}{k^*}\downarrow$: $\frac{f(k^*_{old})}{k^*_{old}}, \frac{f(k^*_{new})}{k^*_{new}}$

$\Rightarrow k^*_{old} < k^*_{new}$ ($k^*\uparrow$)

(i) $s\uparrow$ a little.

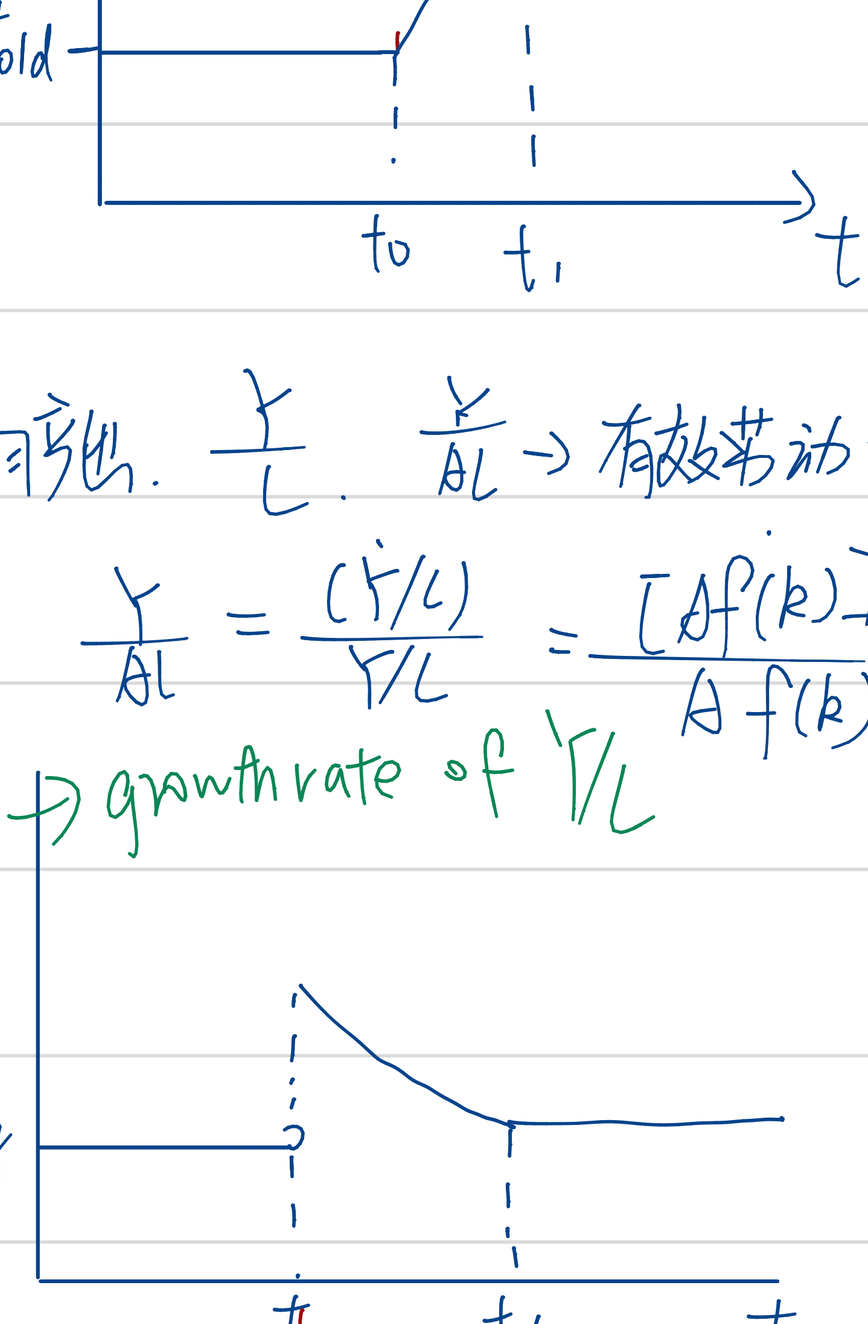
$\dot{k} \to sf(k) - (\delta+n+g)k$.





s的变化只有水平效应，没有替代效应.

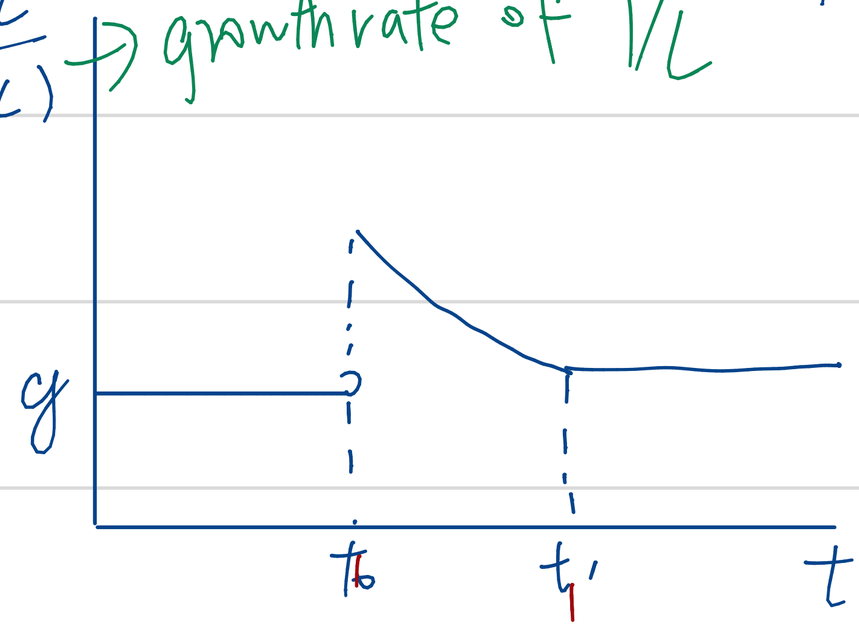
使得k^*上升部分. 但又达到0.



人均资本. $\frac{\dot{Y}}{L}$. $\frac{\dot{Y}}{AL}$ → 有效劳动的人均资本.

$$\frac{\dot{Y}}{AL} = \frac{(\dot{Y}/L)}{Y/L} = \frac{[Af(k)]\dot{}}{Af(k)} = \frac{\dot{A}}{A} + \frac{f(\dot{k})}{f(k)} \qquad k(t)$$

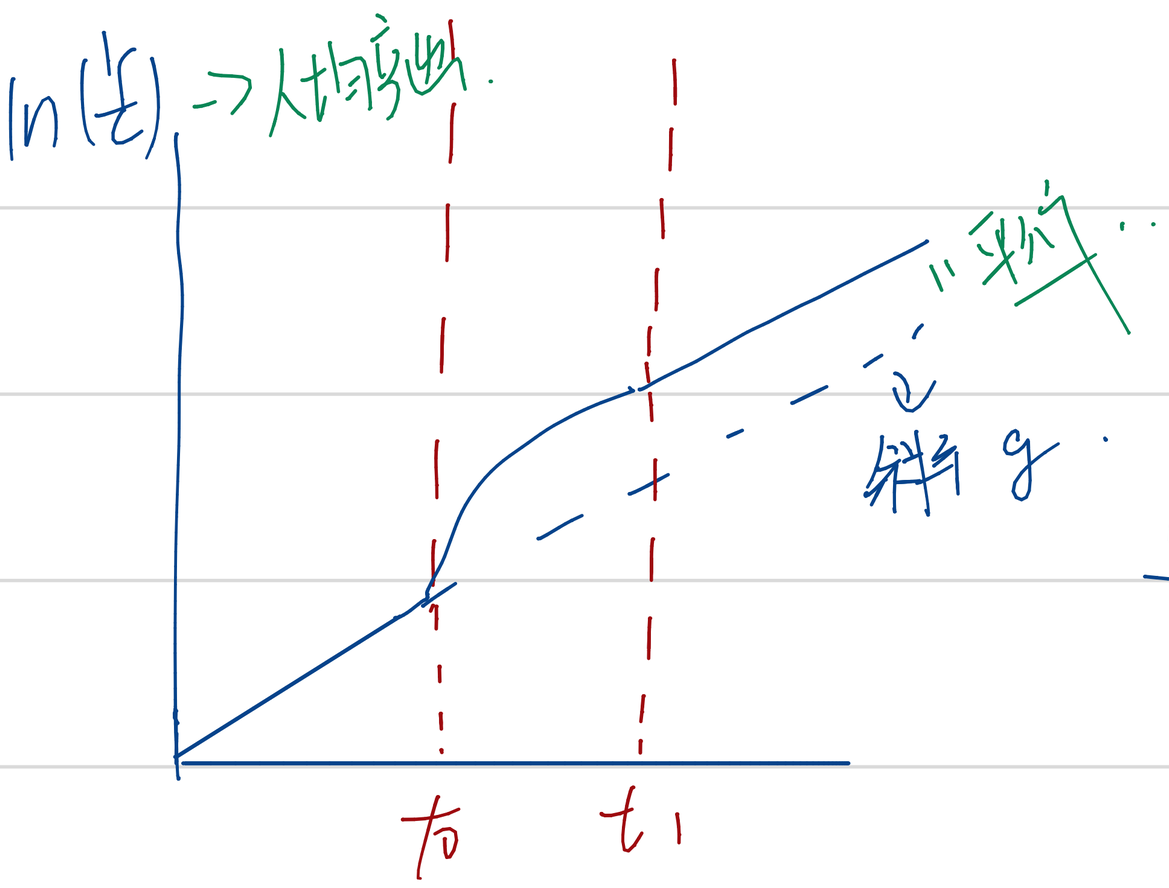
$\frac{\dot{Y}/L}{(Y/L)}$ → growth rate of Y/L



$$= g + \frac{f'(k)}{f(k)}\dot{k}$$

① $\dot{k}\uparrow$ at t_0

② $k\uparrow$ after $t_0 \Rightarrow \dot{k}\downarrow$.



$\frac{f'(k)}{f(k)}$ ↓ · so $\frac{(\dot{Y}/L)}{Y/L}$ ↓

$\frac{d\ln(Y/L)}{dt}$

$= \frac{(\dot{Y}/L)}{Y/L}$

证明: $y^* = f(k^*)$

$\frac{\partial y^*}{\partial s} = f'(k^*) \frac{\partial k^*(s,n,g,\delta)}{\partial s}$ 稳态情况.

$\dot{k}=0$ 均衡点

∴ $sf(k^*(s,n,g,\delta)) = (n+g+\delta)\cdot k^*(s,n,g,\delta)$

用导 s

$sf'(k^*)\frac{\partial k^*}{\partial s} + f(k^*) = (n+g+\delta)\cdot\frac{\partial k^*}{\partial s}$

$\therefore \frac{\partial k^*}{\partial s}[sf'(k^*) - (n+g+\delta)] = -f(k^*)$

$\frac{\partial k^*}{\partial s} = \frac{f(k^*)}{n+g+\delta - sf'(k^*)}$

代入 $\boxed{\frac{\partial y^*}{\partial s}} = f'(k^*)\cdot\frac{f(k^*)}{n+g+\delta-sf'(k^*)}$

弹性 elasticity. 一个变量变化1%. 另一个变化%多少.

eg. 一个国家储蓄率从60%变成 60%×1.01, 产出? a(1+x%)

$\frac{s}{y^*}\cdot\frac{\partial y^*}{\partial s} = \frac{s}{f(k^*)}\cdot f'(k^*)\cdot\frac{f(k^*)}{n+g+\delta-sf'(k^*)}$

$$\therefore s = \frac{(n+g+\delta)k^*}{f(k^*)}$$

$$= \frac{s}{f(k^*)} \cdot f'(k^*) \cdot \frac{f(k^*)}{n+g+\delta - \frac{(n+g+\delta)\cdot k^* \cdot f'(k^*)}{f(k^*)}}$$

$$= \frac{sf'(k^*)}{n+g+\delta - \frac{(n+g+\delta)\cdot k^* \cdot f'(k^*)}{f(k^*)}}$$

$$= \frac{(n+g+\delta)k^* \cdot f'(k^*)}{f(k^*)} \cdot \left(\frac{f(k^*)}{[f(k^*) - k^* f'(k^*)] \cdot (n+g+\delta)} \right)$$

$$= \frac{k^* f'(k^*)}{f(k^*) - k^* f'(k^*)} = \frac{k^* f'(k^*) / f(k^*)}{1 - \frac{k^* f'(k^*)}{f(k^*)}}$$

定义 $a_K(k^*) = k^* \cdot \frac{f'(k^*)}{f(k^*)}$

资本家 现在的 所得

增加一单位资本所 增加的产出 "资本回报率"

总产出 ↓ 100

eg. 生产100. 资本家有60.

$$\frac{s}{y^*}\cdot\frac{\partial y^*}{\partial s}=\frac{a_k(k^*)}{1-a_k(k^*)}$$

↓

Cobb-Douglas 中 a.

正

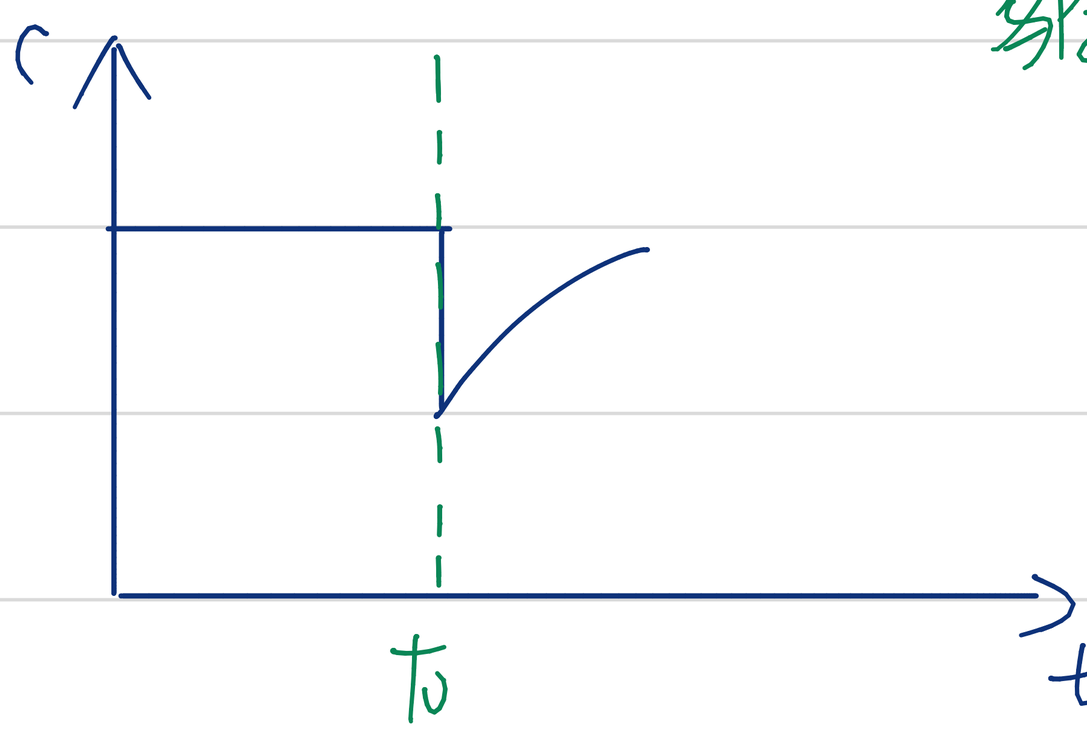
资本收入份额越高个. 提供储蓄对于产出影响更大个

高.

$\therefore a_k$个 $\Rightarrow \frac{s}{y}\cdot\frac{\partial y^*}{\partial s}$个

找 the impact of a change in the s

② 对 consumption



储蓄率. 减少当下消费.

$$C^*=f(k^*)-sf(k^*)$$

$$=f(k^*)-(n+g+\delta)\cdot k^*$$

$$=f(k^*(s,n,g,\delta))-$$

$$(n+g+\delta)\cdot k^*(s,n,g,\delta)$$

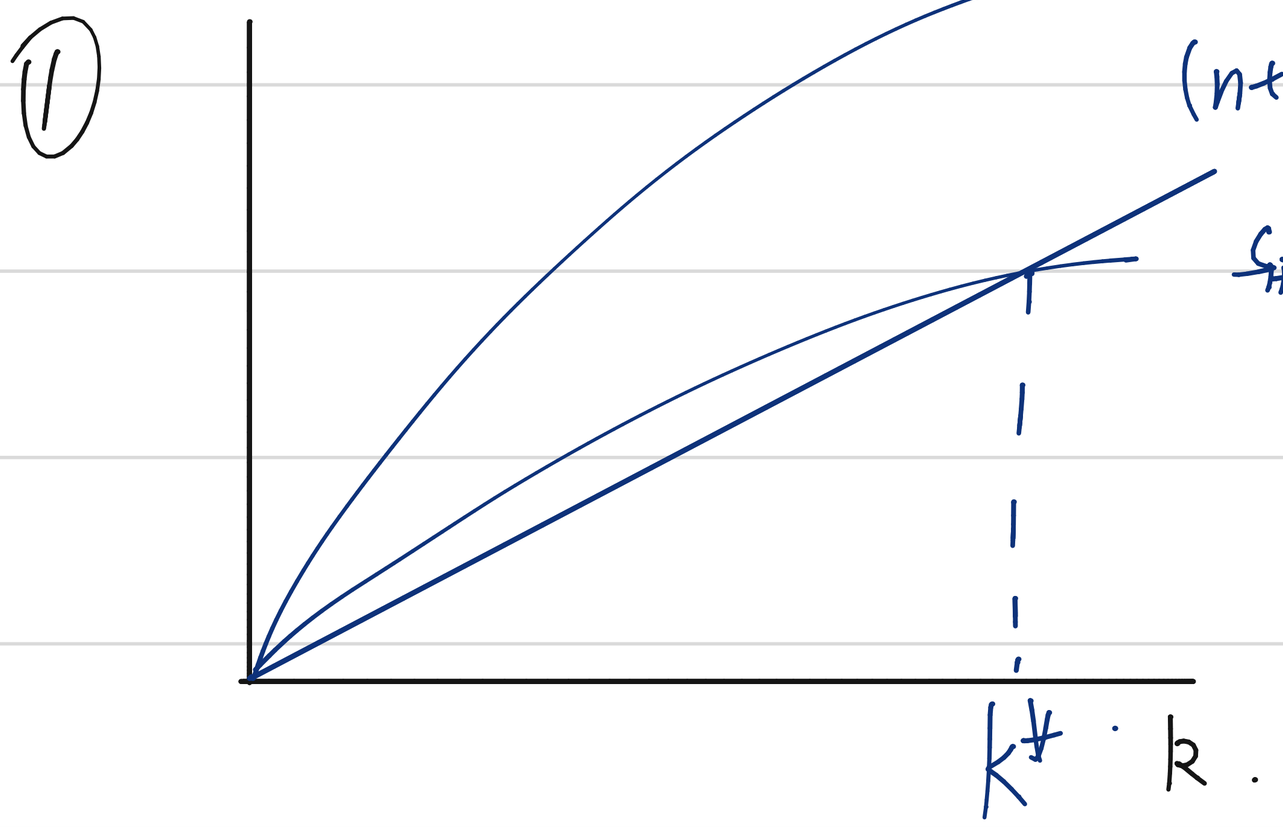
$$\frac{\partial C^*}{\partial s}=f'(k^*)\cdot\frac{\partial k^*}{\partial s}\cdot-(n+g+\delta)\cdot\frac{\partial k^*}{\partial s}$$

$$=\frac{\partial k^*}{\partial s}\cdot\underbrace{(f'(k^*)-(n+g+\delta))}.$$

⇓

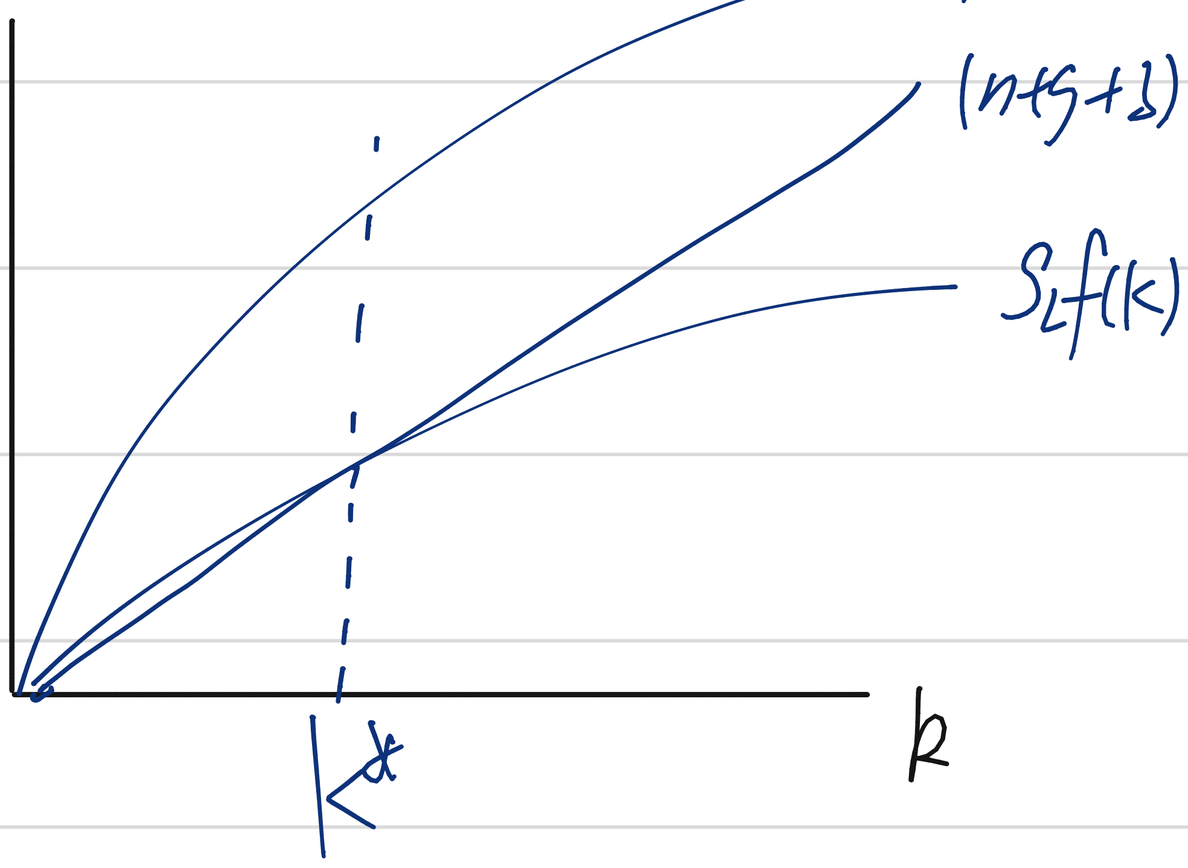
∵ s↑. k↑ ∴ positive $\frac{\partial k^*}{\partial s}$. ↓ 不一定正

3个情况

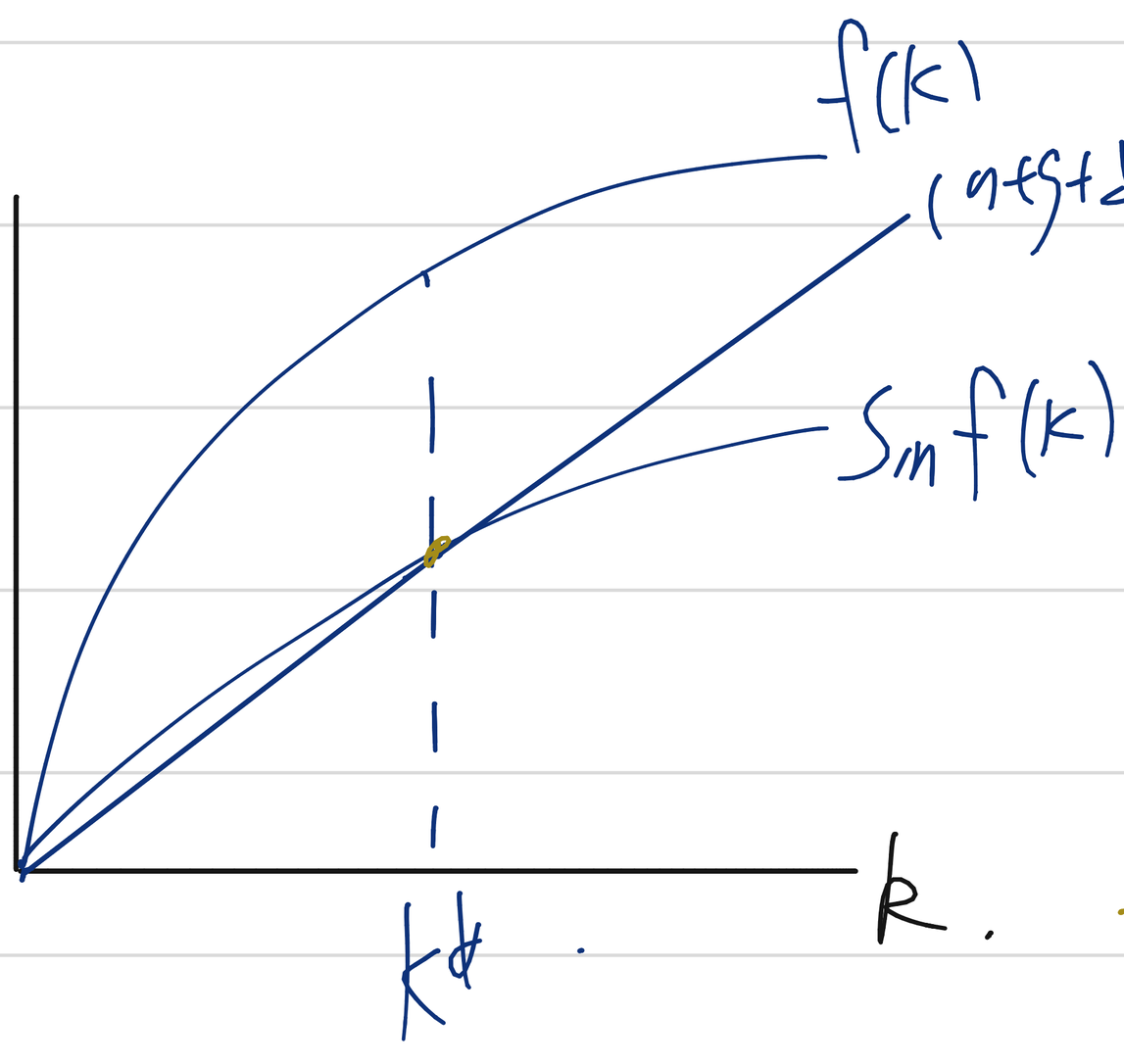
① 

负 好

∴ 当s很高. s个→C↓

② 

当s降. S↑ C↑.

③ 

最大消费

选个 最佳均衡. 稳定.

The golden-rule: $f'(k^*) = (n+g+s)$

s 随时间变化对 c 无影响.

Growth accounting

$$Y(t) = F(K(t), A(t)L(t)) \quad ①$$

$$\dot{Y}(t) = \frac{\partial Y(t)}{\partial K(t)}\cdot\dot{K}(t) + \frac{\partial Y(t)}{\partial A(t)}\cdot\dot{A}(t) + \frac{\partial Y(t)}{\partial L(t)}\cdot\dot{L}(t) \quad ②$$

$$\frac{\partial Y(t)}{\partial L(t)} = \frac{\partial Y(t)}{\partial A(t)\cdot L(t)}\cdot A(t) \qquad \frac{\partial Y(t)}{\partial A(t)} = \frac{\partial Y(t)}{\partial A(t)L(t)}\cdot L(t). \quad 代入②.$$

$$\dot{Y}(t) = \frac{\partial Y(t)}{\partial K(t)}\cdot\dot{K}(t) + \frac{\partial Y(t)}{\partial A(t)L(t)}\cdot L(t)\dot{A}(t) + \frac{\partial Y(t)}{\partial A(t)L(t)}\cdot A(t)\times\dot{L}(t)$$

$$\frac{\dot{Y}(t)}{Y(t)} = \frac{\partial Y(t)}{\partial K(t)}\cdot\frac{\dot{K}(t)}{K(t)}\cdot\frac{K(t)}{Y(t)} + \frac{\partial Y(t)}{\partial A(t)}\cdot\frac{\dot{A}(t)}{A(t)}\cdot\frac{A(t)}{Y(t)} + \frac{\partial Y(t)}{\partial L(t)}\cdot\frac{\dot{L}(t)}{L(t)}\cdot\frac{L(t)}{Y(t)}$$

经济增长率.

已知 $\frac{K(t)}{Y(t)}\cdot\frac{\partial Y(t)}{\partial K(t)}$ 是常数

是资本对设计的弹性.

$$\therefore \alpha_K(t)\cdot\frac{\dot{K}(t)}{K(t)} + \alpha_L(t)\frac{\dot{L}(t)}{L(t)} + R(t).$$

Residual 索洛剩余.

资本增 1%. 产出... 资本贡献

劳动力增加 1%. 产出.. 劳动力贡献

$$\therefore R(t) = \frac{\partial Y(t)\cdot\dot{A}(t)A(t)}{\partial A(t)A(t)Y(t)}.$$

Tutorial 2.

Exercise 1.

(a) Interpret $F_K(K,L)>0$ $F_{KK}(K,L)<0$.

$F_K(K,L)$ is the marginal product of capital, it means when captial increase 1unit, how much the output will increase

So if $F_K(K,L)>0$. thus, captial rise more, the output increase more.

$F_{KK}(K,L)$ is the change of the marginal product of capital, therefore if $F_{KK}(K,L)<0$, it means as the amount of capital increases, the additional output generated by extra unit of capital becomes smaller.

show that F has constant returns to scale and interpret its economic meaning.

We need to prove that: $F(tK, tL; A) = tF(K,L;A)$ Interpret: it means if both capital and labour are doubled. output also doubled.

$$F(tK,tL;A)=\left(a\cdot tK^{\frac{\sigma-1}{\sigma}}+t(1-a)(AL)^{\frac{\sigma-1}{\sigma}}\right)^{\frac{\sigma}{\sigma-1}}$$

Assume $\frac{\sigma-1}{\sigma}=\rho$.

And this model contains no omitted fixed factor

$$\therefore F(tK,tL;A)=\left(a(tK)^{\rho}+(1-a)(tAL)^{\rho}\right)^{1/\rho}$$

$$=\left[a\cdot K^{\rho}\cdot t^{\rho}+(1-a)\cdot t^{\rho}(AL)^{\rho}\right]^{1/\rho}.$$

$$=\left[t^{\rho}\left(a\cdot K^{\rho}+(1-a)(AL)^{\rho}\right)\right]^{1/\rho}$$

$$=t^{\rho\cdot 1/\rho}\left[a\cdot K^{\rho}+(1-a)(AL)^{\rho}\right]^{1/\rho}$$

$$=t\left[a\cdot K^{\rho}+(1-a)(AL)^{\rho}\right]^{1/\rho}.$$

$$=tF(K,L;A)$$

$$\therefore F(tK,tL;A)=tF(K,L;A).$$

(b) Inada condition.

$\lim_{k\to 0} F_k = \infty$ $\lim_{k\to\infty} F_k = 0$ Assume $\frac{\sigma-1}{\sigma} = \rho$.

According to the $F_k(K.L) = a\left(a+(1-a)\left(\frac{AL}{K}\right)^{\rho}\right)^{\frac{1-\rho}{\rho}}$.

$$F_k = a\left(a+(1-a)\left(\frac{AL}{K}\right)^{\rho}\right)^{\frac{1}{\rho}-1}$$

Therefore if $\sigma \in (1,\infty)$, and $k\to 0$, $\therefore \rho > 0 \therefore \left(\frac{AL}{K}\right)^{\rho} \to \infty$ $\frac{1}{\sigma-1} > 0$ $\therefore F_k \to \infty$. ✓

when $k\to\infty$ $\frac{AL}{K}\to 0$. $\because a \cdot a^{\frac{1}{\sigma-1}} = a^{\frac{\sigma}{\sigma-1}}$ $\therefore$ $F_k \to a^{\frac{\sigma}{\sigma-1}} > 0$ ×

if $\sigma \in (0,1)$, and $k\to 0$. $\frac{AL}{K}\to\infty$. $\rho < 0$ $\therefore \left(\frac{AL}{K}\right)^{\rho}\to 0$ $\frac{1}{\sigma-1} < 0$ $\therefore$ $F_k \to a^{\frac{\sigma}{\sigma-1}} < \infty$ ×

$k\to\infty$ $\frac{AL}{K}\to 0$, $\rho < 0$, $\left(\frac{AL}{K}\right)^{\rho}\to\infty$ $\therefore \frac{1}{\sigma-1} < 0 \therefore F_k \to 0$. ✓

Interpret: when $\sigma > 1$, capital and labour are said to be substitutes.

So, if labour is fixed, the production can be continued by increasing capital. So F_k can never fall to zero.

when $\rho < 1$, they are complementary goods. No matter how much capital there is. if no use without sufficient labour to complement it, therefore, when capital is abundant, F_k tends to 0. However, when capital is scarce, F_k not infinite. because of the constraint of labour insufficient.

(c) capital share grow or fall in $K(t)$?

① $\sigma > 1 \to \rho > 0$ $\therefore$ when $K\uparrow \to \frac{AL}{K(t)}\downarrow \therefore \left(\frac{AL}{K(t)}\right)^{\rho}\downarrow \because a$ unchange $\therefore X_{K(t)}\uparrow$.

It means when capital is substitutes, when capital increase, firm will use more capital to replace labour, therefore the capital share increase by capital increasing.

② $\sigma < 1 \to \rho < 0$, when $K\uparrow \to \frac{AL}{K(t)}\downarrow : \left(\frac{AL}{K(t)}\right)^{\rho}\uparrow \because a$ unchange $\therefore X_{K(t)}\downarrow$.

It means capital and labour are complementary; as the amount of capital increase; the price per unit of

capital falls faster than the quantity of capital of capital rises,

so the total share of capital decline.

③ When $\theta=1$ $\therefore \alpha_K = \frac{a}{a+(1-a)}$ constant.

d). show $K^* = AL\left(\frac{1-a}{\left(\frac{\delta}{s}\right)^{\frac{\theta-1}{\theta}}-a}\right)^{\frac{\theta}{\theta-1}}$

$$\dot{k} = -\delta k^* + s\bar{f}(k^*, L; A)$$

$\because$ Steady state of capital exists

$\therefore \dot{k}=0$ $\therefore \delta k^* = s\bar{f}(k^*, L; A)$

$$\delta = \frac{s\bar{f}(k^*, L; A)}{k^*}$$

$$\because f(k, L; A) = \left(a k^\rho + (1-a)(AL)^\rho\right)^{1/\rho}$$

$$= \left[k^\rho\left(a+(1-a)\left(\frac{AL}{k}\right)^\rho\right)\right]^{1/\rho}$$

$$= k\left[a+(1-a)\left(\frac{AL}{k}\right)^\rho\right]^{1/\rho}$$

$$\frac{f(k, L; A)}{k} = \left[a+(1-a)\left(\frac{AL}{k}\right)^\rho\right]^{1/\rho}$$

$$\therefore \delta = s\cdot\left[a+(1-a)\left(\frac{AL}{k^*}\right)^\rho\right]^{1/\rho}.$$

$$\frac{\delta}{s} = \left[a+(1-a)\left(\frac{AL}{k^*}\right)^\rho\right]^{1/\rho}.$$

$$\left(\frac{\delta}{s}\right)^\rho = a+(1-a)\left(\frac{AL}{k^*}\right)^\rho.$$

$$\left(\frac{AL}{k^*}\right)^\rho = \frac{\left(\frac{\delta}{s}\right)^\rho - a}{(1-a)}.$$

$$\frac{AL}{k^*} = \left[\frac{\left(\frac{\delta}{s}\right)^\rho - a}{(1-a)}\right]^{1/\rho}.$$

$$K^* = \frac{AL}{\left[\frac{\left(\frac{\delta}{s}\right)^\rho - a}{1-a}\right]^{1/\rho}} = AL\cdot\left(\frac{1-a}{\left(\frac{\delta}{s}\right)^\rho - a}\right)^{1/\rho} \quad \because \rho = \frac{\theta-1}{\theta}$$

$$\therefore K^* = AL\left(\frac{1-a}{\left(\frac{\delta}{s}\right)^{\frac{\theta-1}{\theta}} a}\right)^{\frac{\theta}{\theta-1}}$$

e) show $g_{K(t)} \equiv \frac{\dot{k}(t)}{k(t)} = s\left(a+(1-a)\left(\frac{AL}{K(t)}\right)^{\frac{\sigma-1}{\sigma}}\right)^{\frac{\sigma}{\sigma-1}} - \delta$

$\dot{k}(t) = sF(K(t), L; A) - \delta k(t)$.

$\frac{\dot{k}(t)}{k(t)} = \frac{sF(k(t), L; A)}{k(t)} - \delta$

$\because \frac{F(k(t))}{k(t)} = \left[a+(1-a)\left(\frac{AL}{k(t)}\right)^{\rho}\right]^{1/\rho}$

$\therefore \frac{\dot{k}(t)}{k(t)} = s\left[a+(1-a)\left(\frac{AL}{k(t)}\right)^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}} - \delta$

f). Assume $\left(\frac{\delta}{s}\right)^{\frac{\sigma-1}{\sigma}} > a \cdot \sigma > 1$

show $k(t) < k^*, g_{K(t)} > 0 \quad k(t) > k^*, g_{K(t)} < 0.$

$\because g_{K(t)} = s\left(a+(1-a)\left(\frac{AL}{k(t)}\right)^{\frac{\sigma-1}{\sigma}}\right)^{\frac{\sigma}{\sigma-1}} - \delta$

when $\sigma > 1 \quad \rho > 0$.

if $g_K > 0$.

$s\left(a+(1-a)\left(\frac{AL}{k(t)}\right)^{\rho}\right)^{1/\rho} - \delta > 0$.

$s\left(a+(1-a)\left(\frac{AL}{k(t)}\right)^{\rho}\right)^{1/\rho} > \delta$

$s^{\rho}\left(a+(1-a)\left(\frac{AL}{k(t)}\right)^{\rho}\right) > \delta^{\rho}$.

$\left(\frac{s}{\delta}\right)^{\rho} < a+(1-a)\left(\frac{AL}{k(t)}\right)$.

$\frac{\left(\frac{s}{\delta}\right)^{\rho} - a}{1-a} < \left(\frac{AL}{k(t)}\right)^{\rho}$.

$\therefore \left[\frac{\left(\frac{s}{\delta}\right)^{\rho} - a}{1-a}\right]^{1/\rho} < \frac{AL}{k(t)}$

$k(t) < AL\left[\frac{1-a}{\left(\frac{s}{\delta}\right)^{\rho} - a}\right]^{1/\rho}$.

$\because \left(\frac{s}{\delta}\right)^{\rho} - a > 0$.

$\therefore k^* = AL\left[\frac{(1-a)}{\left(\frac{s}{\delta}\right)^{\rho} - a}\right]^{1/\rho}$.

$\therefore k(t) < k^*$.

if $g_K < 0$ . . . . $K(t) > K^*$.

② when $b \in (0,1)$.

if $g_{K(t)} > 0$.

$\therefore \left(\frac{s}{\delta}\right)^\rho < a + (1-a)\left(\frac{\Delta L}{K(t)}\right)^\rho$.

$\frac{s}{\delta} < \left[a + (1-a)\left(\frac{\Delta L}{K(t)}\right)^\rho\right]^{1/\rho}$.

$\because b \in (0,1) \quad \therefore \rho < 0$.

$\left(\frac{s}{\delta}\right)^\rho > a + (1-a)\left(\frac{\Delta L}{K(t)}\right)^\rho$

$\frac{\left(\frac{s}{\delta}\right)^\rho - a}{1-a} > \left(\frac{\Delta L}{K(t)}\right)^\rho$.

$\left[\frac{\left(\frac{s}{\delta}\right)^\rho - a}{1-a}\right]^{1/\rho} < \frac{\Delta L}{K(t)}$

$\frac{1}{K(t)} > \frac{1}{\Delta L}\left(\frac{\left(\frac{s}{\delta}\right)^\rho - a}{1-a}\right)^{1/\rho}$.

$K(t) < \Delta L\left[\frac{1-a}{\left(\frac{s}{\delta}\right)^\rho - a}\right]^\rho$.

$\because K^* = \Delta L\left[\frac{1-a}{\left(\frac{s}{\delta}\right)^\rho - a}\right]^\rho$.

$\therefore K(t) < K^*$. ~~→~~ $g_{K(t)} < 0 \quad K(t) > K^*$

(4). $(\frac{\delta}{s})^{\frac{\theta-1}{\theta}} \leq a$., $\theta > 1$ ① show $g_k(t)$ always positive.

② level of k as $t \to \infty$, ③ steady state exist? ④ output? ⑤ intuition

$\therefore g_k = s[a + (1-a)(\frac{AL}{K})^{\rho}]^{1/\rho} - \delta$ $\because \theta > 1 \therefore \rho > 0$.

$\because (1-a) > 0$ $(\frac{AL}{K}) > 0$, $\rho > 0$

$\therefore a + (1-a)(\frac{AL}{K})^{\rho} > a$

$[a + (1-a)(\frac{AL}{K})^{\rho}]^{1/\rho} > a^{1/\rho}$ $\because s > 0$

$s[a + (1-a)(\frac{AL}{K})^{\rho}]^{1/\rho} > sa^{1/\rho}$

$s[a + (1-a)(\frac{AL}{K})^{\rho}]^{1/\rho} - \delta > sa^{1/\rho} - \delta$.

$\therefore 1 > (\frac{\delta}{s})^{\rho} \leq a$.

$\frac{\delta}{s} \leq a^{1/\rho}$

$\delta \leq sa^{1/\rho}$.

$\therefore sa^{1/\rho} - \delta \geq 0$.

$\therefore s[a + (1-a)(\frac{AL}{K})^{\rho}]^{1/\rho} > 0$.

② $\because g_k > 0$ $\therefore k(t)$ increasing.

$\therefore t \to \infty$ $k(t) \to \infty$.

When $k \to \infty$. $\frac{AL}{K} \to 0$. $a > 0$ $s > 0$

$\therefore g_k \to sa^{1/\rho} - \delta > 0$.

$\therefore$ if $(\frac{\delta}{s})^{\rho} < a$. $\therefore g_k$ is positive and increase continuously, $k \to \infty$.

if $(\frac{\delta}{s})^{\rho} = a$. $\therefore g_k \to 0$, but $k \to \infty$.

③ There is no steady, because steady need $g_K = 0$ but $g_K > 0$.

④ output

from d) $\frac{Y}{K} = [a + (1-a)(\frac{AL}{K})^\rho]^{1/\rho}$

$\because K \to \infty$. $\frac{AL}{K} \to 0$.

$\therefore \frac{Y}{K} \to a^{1/\rho}$.

$\therefore Y \to k \cdot a^{1/\rho}$.

⑤ $\sigma > 1$, substitute $\to K \to \infty$. marginal product of capital not fall to 0, to $a^{1/\rho}$

And $s a^{1/\rho} \geq \delta$. if s is enough large.

the investment increased by per capital can more than δk.

so capital increase.

b) if $\sigma \in (0,1)$. $\rho < 0$.

① $\because a + (1-a)(\frac{AL}{K})^\rho > a$.

$[a + (1-a)(\frac{AL}{K})^\rho]^{1/\rho} < a^{1/\rho}$.

$s[a + (1-a)(\frac{AL}{K})^\rho]^{1/\rho} < s a^{1/\rho}$.

$s[a + (1-a)(\frac{AL}{K})^\rho]^{1/\rho} - \delta < s a^{1/\rho} - \delta$

$\because (\frac{\delta}{s})^\rho \leq a$.

$\frac{\delta}{s} \geq a^{1/\rho}$.

$\delta \geq sa^{1/\rho}$. $\therefore sa^{1/\rho} - \delta \leq 0$.

$\therefore s\left[a + (1-a)\left(\frac{\Delta L}{K}\right)^{\rho}\right]^{1/\rho} < 0$.

$\therefore g_K < 0$.

② K decreasing.

$\because g_K < 0$. $\therefore t \to \infty$ $K(t) \to 0$.

③ no exist steady state because $g_K < 0$.

output: $\because Y/K \to a^{1/\rho}$.

$\therefore Y \to a^{1/\rho} K$.

$\because K \to 0$.

$\therefore Y \to 0$.

④ $\because \delta \geq sa^{1/\rho}$ means investment less than δk (折旧)